

Database Description for Selected Tables

CD

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, same as ID of DiscProduct of which type is CD
2.			artist	VARCHAR(100)	Yes	Name of the artist or performer
3.			recordLabel	VARCHAR(100)	Yes	Record label of the CD
4.			track	TEXT	Yes	List of tracks in the CD

DVD

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, same as ID of DiscProduct of which type is DVD
2.			discType	VARCHAR(45)	Yes	Disc type of the DVD
3.			director	VARCHAR(100)	Yes	Director of the DVD
4.			studio	VARCHAR(100)	Yes	Production studio or manufacturer
5.			subtitles	VARCHAR(255)	Yes	Available subtitles of the DVD

DiscProduct

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, same as ID of Product of which type is DiscProduct
2.			releaseDate	DATE	Yes	Release date of the disc product
3.			genre	VARCHAR(100)	Yes	Genre of the disc product
4.			language	VARCHAR(45)	Yes	Language of the disc product
5.			totalLength	INTEGER	Yes	Total length of the disc product in minutes

Invoice

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, auto increment
2.			VATsubtotal	NUMERIC(12,2)	Yes	VAT subtotal of the invoice

3.			totalAmount	NUMERIC(12,2)	Yes	Final total amount of the invoice
4.		x	orderId	BIGINT	Yes	Order ID associated with this invoice

Transaction

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, auto increment
2.			amount	NUMERIC(12,2)	Yes	Paid transaction amount
3.			content	TEXT	Yes	Transaction content or payment note
4.			method	VARCHAR(45)	Yes	Payment method, e.g., credit card, VietQR
5.			status	VARCHAR(30)	Yes	Transaction status, e.g., pending, success, failed
6.		x	invoiceId	BIGINT	Yes	Invoice ID associated with this transaction

User

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, auto increment
2.			userName	VARCHAR(100)	Yes	Username used for authentication
3.			email	VARCHAR(255)	Yes	User email address
4.			hashedPass	VARCHAR(255)	Yes	Hashed password of the user
5.			status	VARCHAR(30)	Yes	User account status, e.g., active, inactive, blocked

Role

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, auto increment
2.			roleName	VARCHAR(100)	Yes	Name of the role assigned to users

UserRole

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x	x	userId	BIGINT	Yes	User ID, foreign key referencing "User"(id)
2.	x	x	roleId	BIGINT	Yes	Role ID, foreign key referencing Role(id)

Product

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	BIGSERIAL	Yes	Product ID, auto increment
2			barcode	VARCHAR(50)	Yes	Product barcode
3			category	VARCHAR(50)	Yes	Product category
4			title	VARCHAR(150)	Yes	Product title
5			description	TEXT	Yes	Product description
6			dimensions	VARCHAR(100)	No	Product dimensions
7			weight	NUMERIC(10,2)	Yes	Product weight
8			originalValue	NUMERIC(12,2)	Yes	Original value of product
9			currentPrice	NUMERIC(12,2)	Yes	Current selling price
10			quantity	INTEGER	Yes	Number of products in stock
11			status	VARCHAR(30)	Yes	Product status
12			image_url	TEXT	Yes	Product image URL
13			video_url	TEXT	Yes	Product video URL
14			createdAt	TIMESTAMP	Yes	Created date
15			updatedAt	TIMESTAMP	No	Updated date

Printable Product

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	BIGSERIAL	Yes	ID
2			publisher	VARCHAR(150)	Yes	Publisher name
3			language	VARCHAR(50)	Yes	Publication language
4			publishDate	DATE	Yes	Publication date

Book

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	BIGSERIAL	Yes	ID
2			coverType	VARCHAR(50)	Yes	Book cover type
3			nbPages	INTEGER	Yes	Number of pages
4			genre	VARCHAR(100)	Yes	Book genre

Newspaper

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	BIGSERIAL	Yes	ID
2			editorInChief	VARCHAR(100)	Yes	Editor-in-chief name
3			issueNumber	VARCHAR(50)	Yes	Newspaper issue number

4			publicationFreq	VARCHAR(50)	Yes	Publication frequency
5			ISSN	VARCHAR(20)	Yes	International Standard Serial Number
6			sections	TEXT	Yes	Newspaper sections

Authors

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	BIGSERIAL	Yes	Author ID, auto increment
2			name	VARCHAR(100)	Yes	Author name

Book_Author

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	bookId	BIGINT	Yes	ID of book
2	x	x	authorId	BIGINT	Yes	ID of author

Orders

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		id	BIGSERIAL	Yes	Order ID, auto increment
2			orderId	VARCHAR(50)	Yes	Order code
3			customerName	VARCHAR(100)	Yes	Customer name
4			phoneNumber	VARCHAR(20)	Yes	Customer phone number
5			email	VARCHAR(150)	Yes	Customer email
6			streetAddress	TEXT	Yes	Customer street address
7			province	VARCHAR(100)	Yes	Customer province
8			deliveryMethod	VARCHAR(50)	Yes	Delivery method
9			deliveryFee	NUMERIC(12,2)	Yes	Delivery fee
10			subtotal	NUMERIC(12,2)	Yes	Order subtotal
11			status	VARCHAR(30)	Yes	Order status
12			createdAt	TIMESTAMP	Yes	Created date
13			updatedAt	TIMESTAMP	No	Updated date

Order_product

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x	x	orderId	BIGINT	Yes	ID of order
2	x	x	productId	BIGINT	Yes	ID of product

3			quantity	INTEGER	Yes	Quantity of product in order
4			price	NUMERIC(12,2)	Yes	Product price at order time

ProductLog

#	PK	FK	Column Name	Data type	Mandatory	Description
1.	x		id	BIGSERIAL	Yes	ID, auto increment
2.			action	VARCHAR(100)	Yes	Action performed on the product, e.g., create, update, delete
3.			createdDate	TIMESTAMP WITH TIME ZONE	Yes	Date and time when the log record is created
4.		x	productId	BIGINT	Yes	Product ID associated with this log
5.		x	createdByUserId	BIGINT	Yes	User ID of the person who created this log